

RESEARCH PROJECT

Momentum Smearing with Machine Learning

Conditional Normalising Flows for Muon Momentum Reconstruction at LHCb

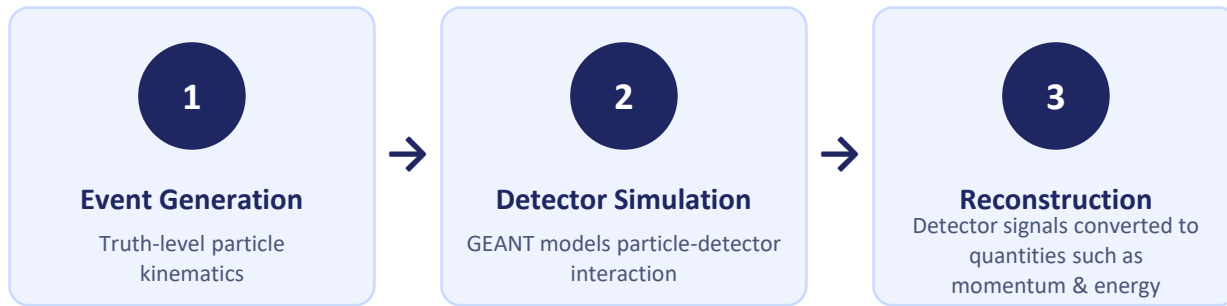
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How do detector simulations work?

A standard HEP analysis involves Monte Carlo (MC) simulations to model physical processes and interpret experimental results. These simulations involve the following stages:

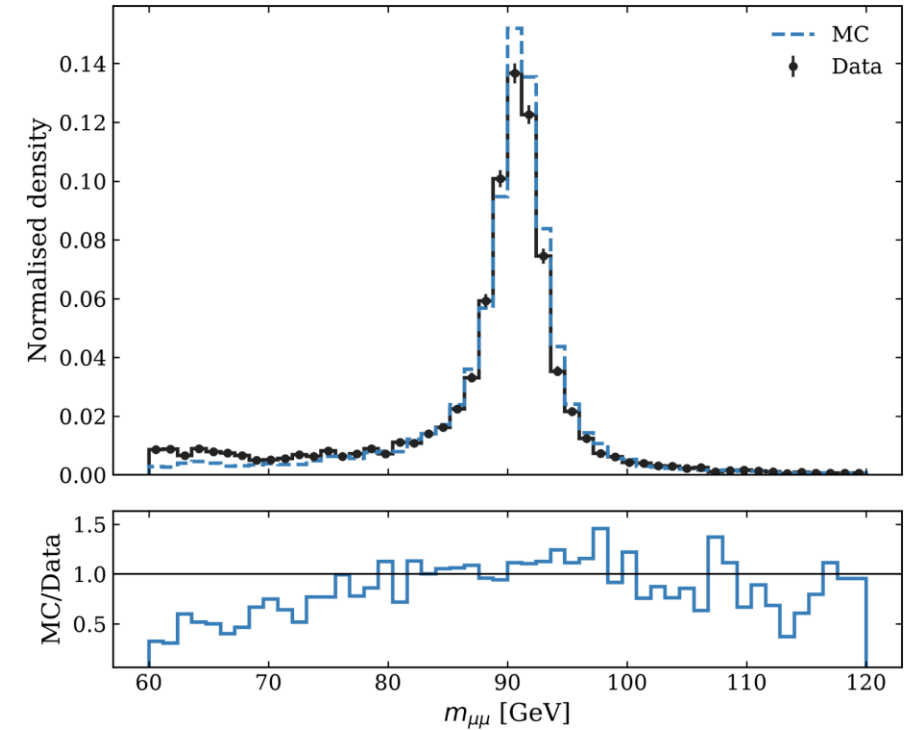


The Problem

Detector effects, such as residual detector misalignment, imperfect material modeling, and finite resolution and granularity mean simulated and real momenta never quite agree.

$$p \rightarrow (1 + \alpha + \beta/p - q\delta p) p$$

Ref. R. Aaij, A.S.W. Abdelmotteleb et al. 2025, Measurement of the Z-boson mass
 Ref. R. Aaij, A.S.W. Abdelmotteleb et al. 2021, Measurement of the W-boson mass



Reco $\neq$ Truth

Finite resolution & imperfect calibration mean MC never perfectly matches data

The Fix

Momentum Smearing

Reconstructed MC momenta are artificially distorted to resemble the resolution effects seen in real detector data.

Traditional smearing uses hand-crafted, parametric functions (e.g. Gaussian noise) tuned against variables like momentum p , pseudorapidity η and charge q . While effective, it is

- Inflexible and hard to tune.
- Other hidden parameters or unknown detector effects.
- Loss of correlation.
- Non-Gaussianity

MOTIVATION

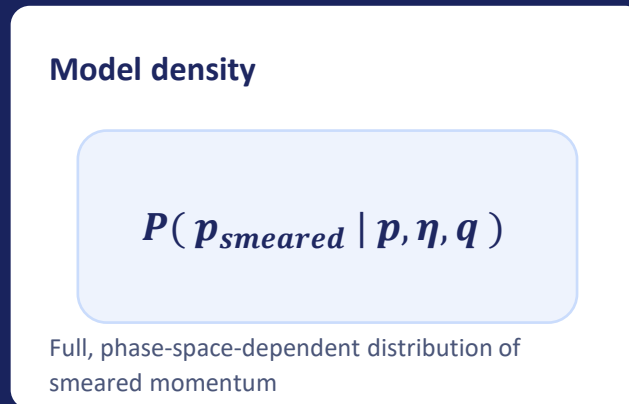
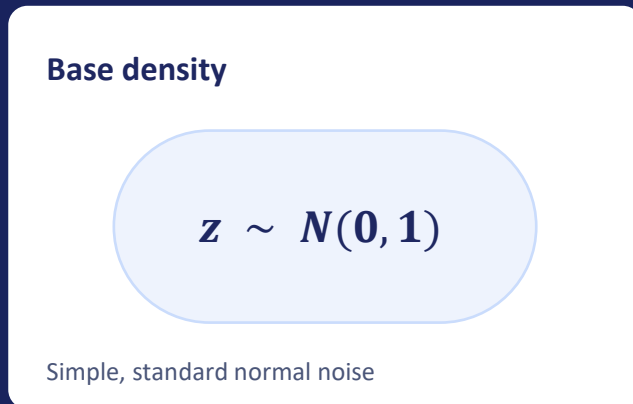
Can data-driven, flexible generative models, such as Conditional Normalising Flows, learn detector smearing directly from data without hand-crafted assumptions?

APPROACH

Conditional Normalising Flows in one picture

A normalising flow learns an invertible, differentiable mapping that transforms simple noise into a complex target distribution.

$$f(z | p, \eta, q)$$

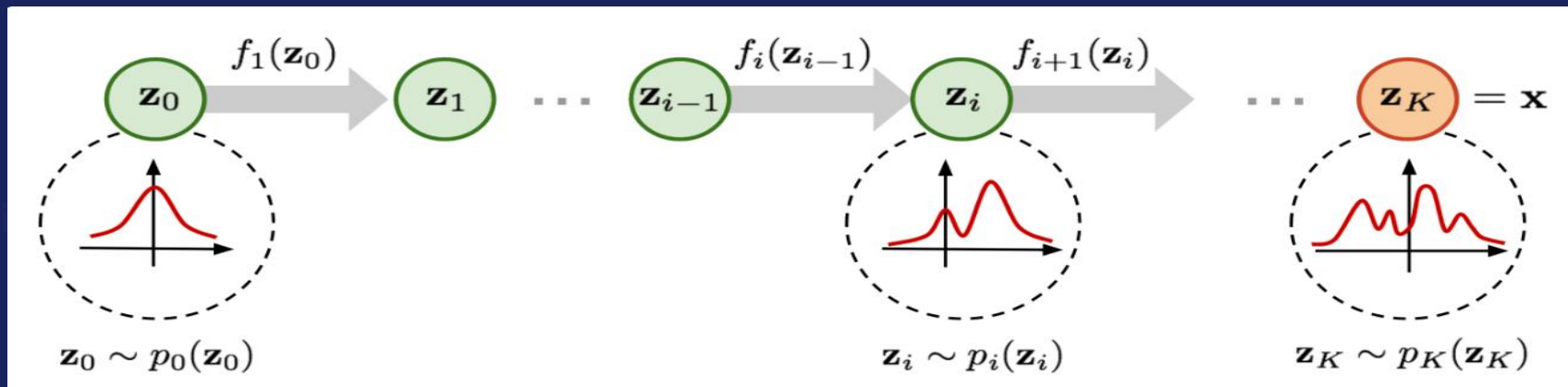


invertible, differentiable
transform

Key difference from regression:

A CNF doesn't predict one smeared momentum, it learns the full conditional distribution, so we can sample stochastically and model uncertainty and correlations.

<https://github.com/bayesiains/nflows>



<https://lilianweng.github.io/posts/2018-10-13-flow-models/>

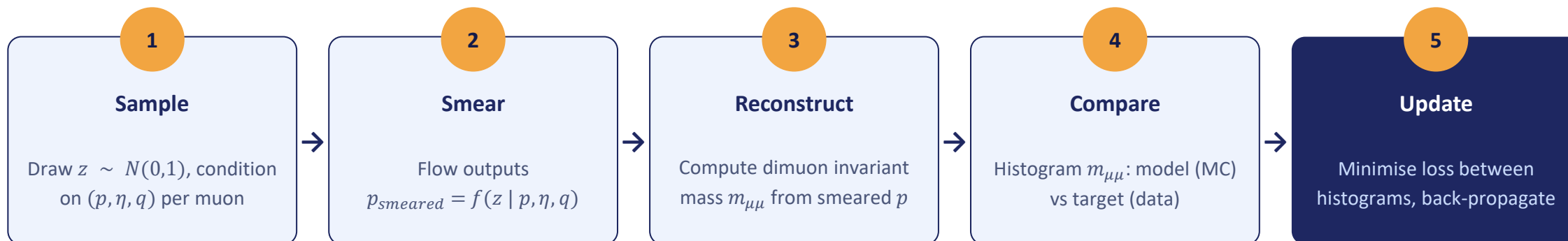
Physics-inspired vs. ML-based smearing

Aspect	Physics-inspired smearing	ML-based smearing (this project)
Knowledge source	Detector physics & parameterisations	Patterns learned from data / simulation
Functional form	Predefined (e.g. Gaussian)	Learned via normalising flow
Correlations	Hard to include	Naturally captured
Flexibility	Limited to assumed model	Very high
Best use case	Quick studies, well-understood effects	Detailed simulation, hard-to-model effects
Validation	Straightforward	Can be complex

***Advantage of ML smearing: high modelling power for realistic detector effects beyond simple assumptions.
Trade-off: less interpretable, needs careful regularisation to avoid overfitting.***

Training: Match invariant mass

The flow is trained so that reconstructed dimuon invariant mass spectra from smeared MC match real data.



Dimuon invariant mass

$$m_{\mu\mu} = \sqrt{2m_\mu^2 + 2[E_1 E_2 - p_1 p_2 \cos\theta]}$$

with $E_i = \sqrt{(m_\mu^2 + p_i^2)}$, p_1, p_2 the muon momenta, θ the angle between them

Loss function (test statistic)

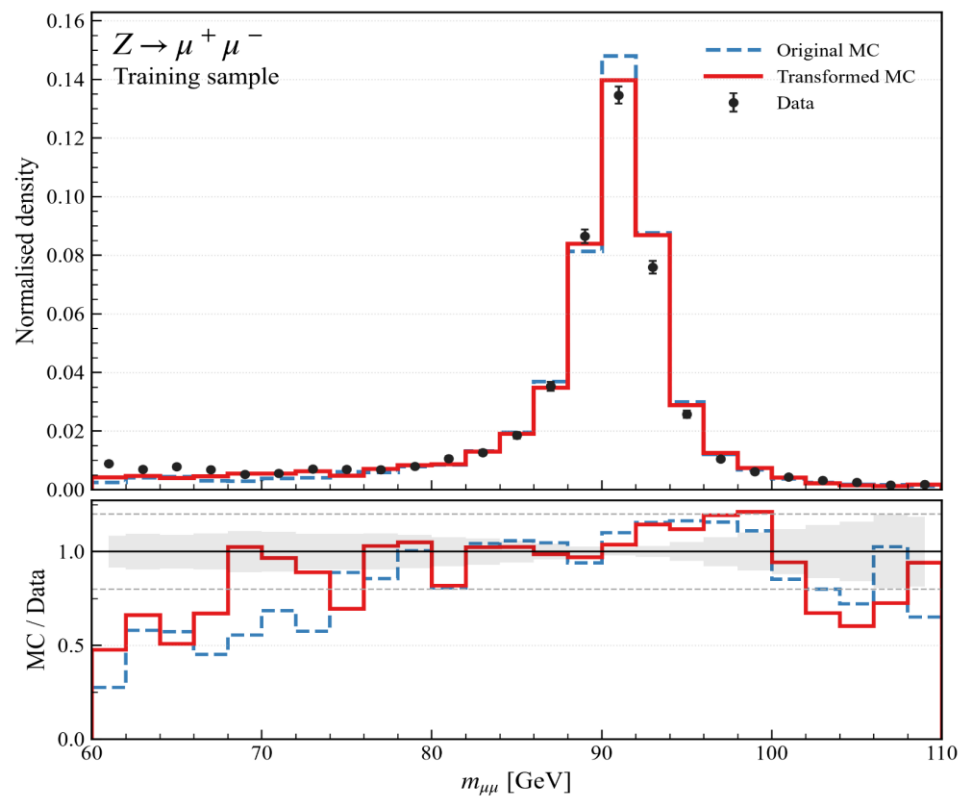
$$L = \sum_i (H_{\text{pred},i} - H_{\text{data},i})^2$$

Binned MSE between predicted and data mass histograms, extensible to χ^2 or negative log-likelihood, provided the statistic stays differentiable.

Results

n_events = 10,000
n_steps = 10,000

Fig. 1 Momentum smearing
(Training Data)



Data training mean: 88.042 GeV

Original MC training mean: 85.533 GeV

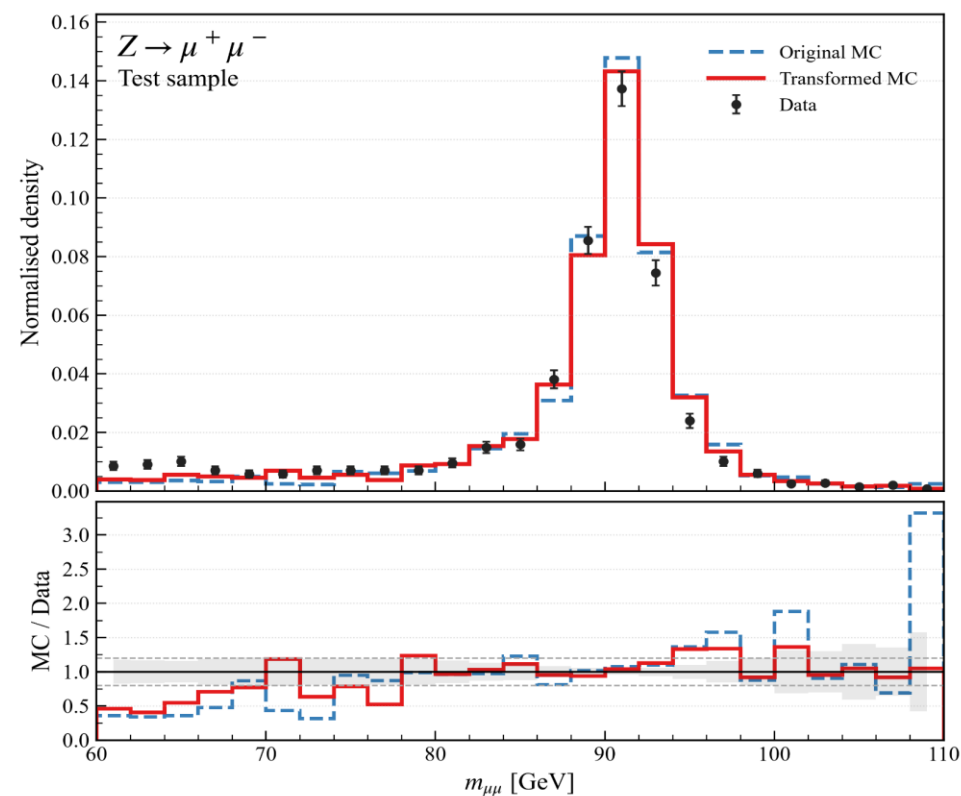
Transformed MC training mean: 87.283 GeV

GeV

$$\chi^2_{MC} = 13.01$$

$$\chi^2_{transformed} = 7.21$$

Fig. 2 Momentum smearing
(Test Data)



Data test mean: 87.704 GeV

Original MC test mean: 85.509 GeV

Transformed MC test mean: 87.065 GeV

$$\chi^2_{MC} = 5.56$$

$$\chi^2_{transformed} = 2.95$$

Results

Fig. 3 Loss

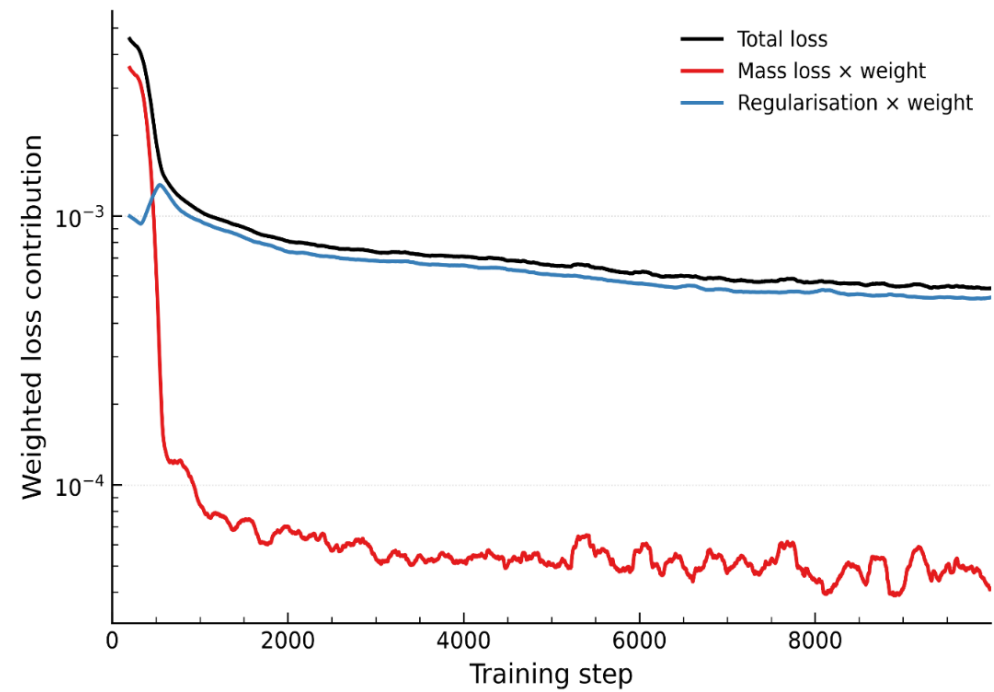
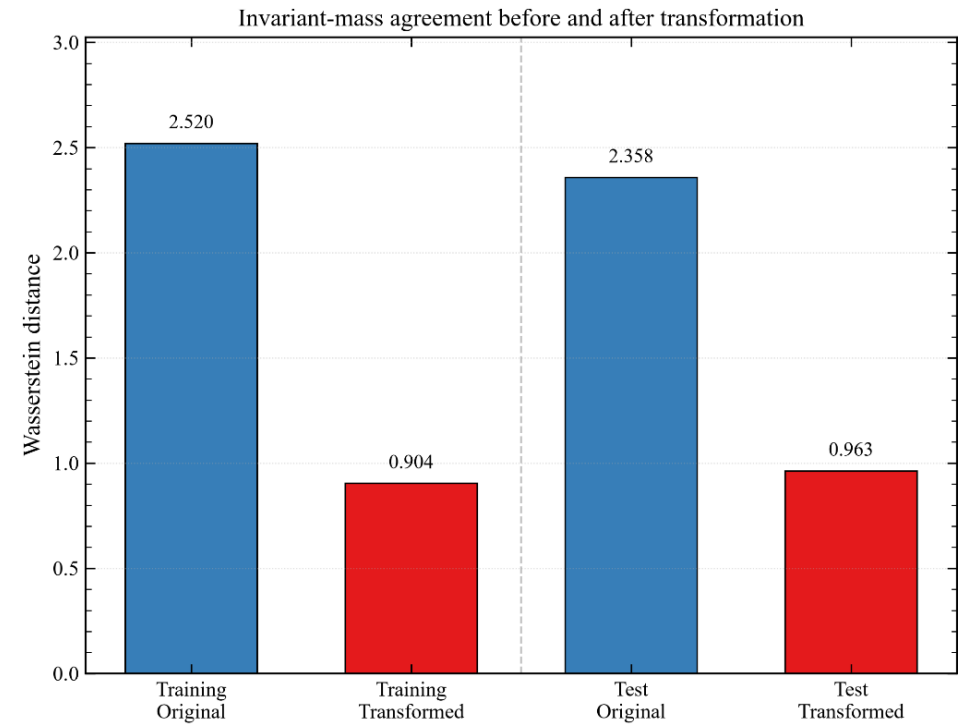


Fig. 4 Improvement



Effectiveness of CNFs

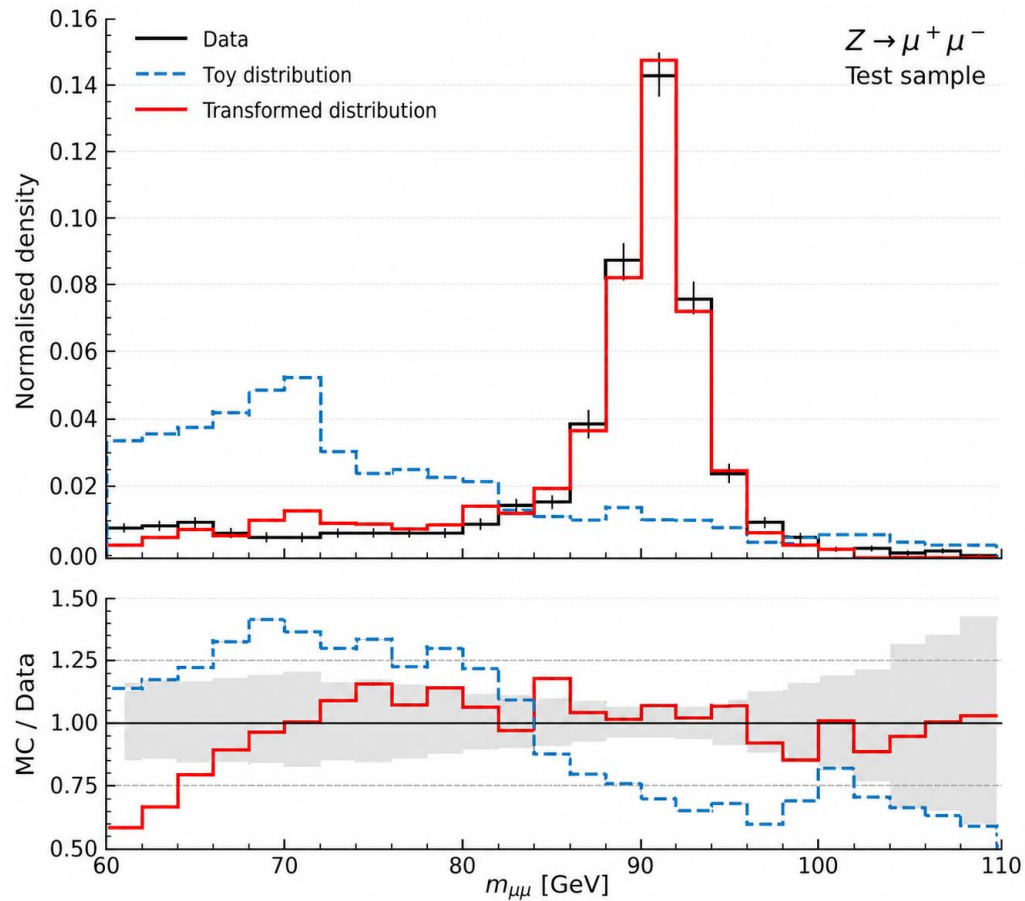


Fig. 5 Smearing a toy distribution

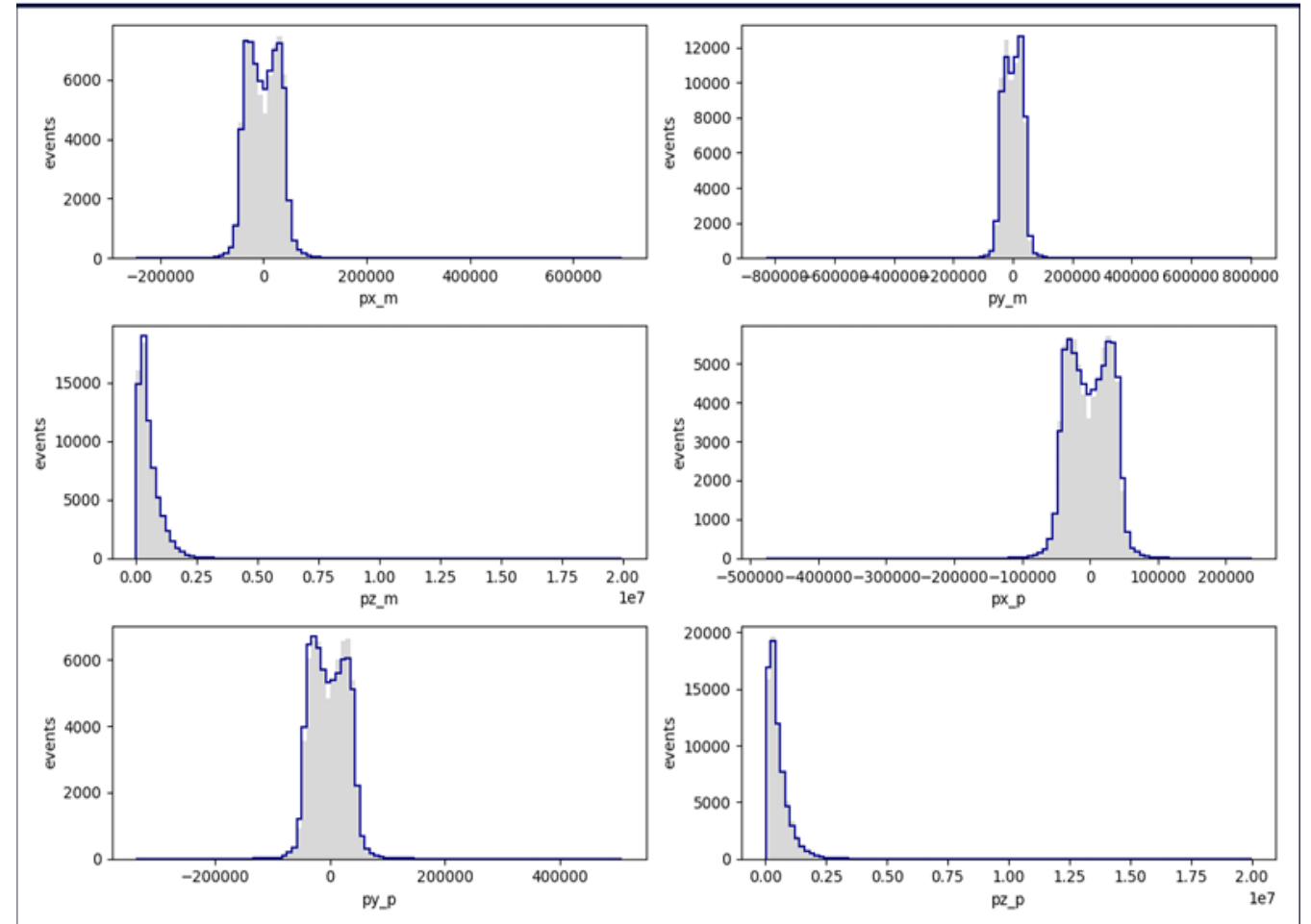


Fig. 6 Plot of individual momenta components

Model Comparison

Model	What is different about it?	Relevance to project
Masked Affine Autoregressive Transform	Transforms each variable one after another using a simple shift and rescaling	Can a relatively simple transformation, while accounting for dependencies between variables, reproduce the data?
Masked Piecewise Rational Quadratic Autoregressive Transform	Transforms each variable one after another using a highly flexible nonlinear function	Does modelling complicated nonlinear changes improve the transformation?
Affine Coupling Transform	Transforms some variables while using the remaining variables as a reference	Can the MC-to-data transformation be learned efficiently without transforming all variables simultaneously?
Piecewise Rational Quadratic Coupling Transform	Uses the same reference-based approach, but allows complicated nonlinear changes	Does combining this efficient approach with a flexible nonlinear transformation give the best result?

Why a flow-based approach pays off



Adapts across phase space

Smearing responds automatically to (p, η, q) without hand-tuned parameterisations for each region.



Full distributions, not points

Sampling instead of point prediction models uncertainty and correlations that a single-value regression can't.



Resample on demand

Draw as many samples as needed for reweighting, pseudo-experiments, or systematic-uncertainty studies.



End-to-end differentiable

Invertible, differentiable transform. Trainable directly against the physics observable that matters, not a proxy.

Net effect: smeared MC that tracks data across the full invariant-mass spectrum, with a model that scales to more conditioning variables (φ , track quality) later.

A data-driven path to better detector simulation

Natural next steps

- Extend conditioning to φ and track-quality variables
- Systematic validation against additional resonances beyond $Z / J/\psi$
- Explore uncertainty estimation enabled by the flow's full conditional distribution
- Test with different batch sizes, number of steps, and number of events

Skills built & how this connects to a PhD

Technical toolkit

- UPROOT for reading MC / data ROOT tuples
- Nflows library for training models and algorithms
- PyTorch for building & training normalising flows
- NumPy / matplotlib for kinematic analysis & plotting
- Differentiable, histogram-based loss design (χ^2 , MSE, NLL)
- Statistical Analysis
- Understanding particle physics jargon and correlation between variables

Why it matters for my PhD

- First hands-on experience turning a generative-ML idea into a working, physics-validated pipeline
- Direct exposure to LHCb data/MC workflows, under supervision from Dr. Mika Vesterinen
- Collaborative research practice with Ahmed Abdelmotteleb — shared codebase, shared debugging
- Concrete bridge between my QFT/particle-physics background and modern ML-for-HEP methods I want to pursue further

The slide features a dark blue background with a central rounded rectangle containing the text "Thank You!". The rectangle has a thin orange border. Scattered around the rectangle are approximately 15 small orange diamond shapes, some pointing up, some down, and some rotated at 45 degrees.

Thank You!